# Real Time Shadow Remover from Images

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***Abstract*—Shadows in images can reduce visual quality, hinder object detection, and affect analysis accuracy. This project presents a real time shadow removal technique using a U Net based deep learning approach. The initial method involved HSV color space conversion and intensity thresholding to detect shadows, followed by brightness compensation. However, limitations in structural restoration led to the adoption of U Net, which excels in image segmentation. The U Net model effectively detects, masks, and removes shadows while preserving textures. We trained the model using the ISTD dataset, incorporating normalization, data augmentation, and a combination of L1 and perceptual loss functions. Experimental results demonstrate improved image clarity and structure retention. Future enhancements include Attention U Net and GAN based refinements.**

***Index Terms*—Shadow Removal, Image Processing, Deep Learning, Convolutional Neural Networks, U-Net Architecture, HSV Color Space, Shadow Masking, Image Segmentation, TensorFlow, PyTorch, Real-Time Implementation, Feature Extraction, Computer Vision, Perceptual Loss, ISTD Dataset.**

## I. Introduction

Shadows are a natural phenomenon caused by the absence or reduction of light due to an object obstructing the light source. While shadows provide important information about object shape and depth, they often degrade image quality and hinder accurate analysis in computer vision tasks. In applications such as surveillance, autonomous driving, and medical imaging, the presence of shadows can lead to incorrect object detection, misclassification, and inaccurate scene interpretation. For instance, in traffic monitoring systems, shadows cast by vehicles can be misinterpreted as new objects, leading to false detection. Similarly, in agricultural monitoring through drone imagery, shadows from trees can obscure vital crop information. Therefore, effective shadow removal is essential to enhance image quality and improve the performance of machine learning models.

Traditional methods for shadow removal relied on intensity thresholding, background subtraction, and color constancy. These approaches primarily focused on identifying shadow regions based on brightness differences and compensating for the lost intensity. However, these methods often failed to handle complex lighting conditions, leading to incomplete shadow removal and texture distortion.

With the emergence of deep learning, Convolutional Neural Networks (CNNs) have shown significant improvements in image segmentation and enhancement tasks. The U-Net architecture, initially developed for biomedical image segmentation, has proven to be highly effective in handling pixel-level tasks, including shadow removal. U-Net's ability to capture both local and global features through encoder-decoder architecture and skip connections makes it an ideal choice for this task.

In this paper, we present a novel shadow removal approach that leverages HSV-based shadow detection and U-Net-based removal. The initial HSV-based method helps in accurately detecting shadow regions by converting the image to the HSV color space and applying intensity thresholding on the Value (V) channel. This creates a shadow mask, which is then refined through the U-Net model to remove the shadows while preserving texture and structure.

The proposed method is evaluated on the ISTD dataset, a standard benchmark for shadow removal, containing 1870 image triplets with shadow, shadow mask, and shadow-free images. The model is trained using a combination of L1 loss and perceptual loss to ensure better texture recovery. Data augmentation techniques and batch normalization are incorporated to enhance the model's generalization capability.

The results demonstrate significant improvement in shadow removal and texture restoration compared to traditional methods. The model achieves high performance in terms of Mean Absolute Error (MAE) and Structural Similarity Index (SSIM). Moreover, the use of Attention U-Net and GAN-based techniques in future work can further refine the output and address

challenges like handling complex shadow boundaries.

This paper is structured as follows: Section II provides background and related work. Section III explains the methodology, including the initial HSV-based approach and the U-Net architecture. Section IV presents the dataset and preprocessing steps. Section V discusses the results and analysis, followed by challenges and future improvements in Section VI. The paper concludes with Section VII.

## II. Background and Related Work

Shadow removal has been a long-standing challenge in the field of computer vision. The presence of shadows can lead to reduced image quality, incorrect object detection, and poor scene understanding. Researchers have explored various methods over the years to tackle this problem.

Early approaches relied on physical models and heuristic-based techniques. Intensity thresholding and chromaticity analysis were some of the first attempts to identify shadow regions by detecting intensity differences between shadowed and non-shadowed areas. However, these methods often struggled with uneven lighting conditions and complex textures.

The introduction of color constancy-based methods improved shadow detection by analyzing the relationship between color and light. Finlayson et al. proposed the use of invariant color models to remove shadows, which worked well under controlled environments but failed in dynamic and outdoor settings.

With the rise of machine learning, data-driven approaches became popular. Support Vector Machines (SVM) and Random Forest classifiers were employed to distinguish shadow and non-shadow pixels. However, these models required extensive feature engineering and lacked the ability to generalize across different datasets.

Deep learning revolutionized the field with the introduction of Convolutional Neural Networks (CNNs). U-Net, originally designed for biomedical image segmentation, gained attention for its ability to capture both local and global features through an encoder-decoder structure with skip connections. This architecture enables the model to accurately detect shadow regions and restore lost texture.

Conditional Generative Adversarial Networks (cGANs) further enhanced the performance by generating high-quality shadow-free images. The adversarial training mechanism allowed the model to produce realistic results. Attention U-Net improved feature selection by focusing on critical regions of the image.

Despite these advancements, challenges like handling complex shadow boundaries, preserving fine details, and adapting to diverse lighting conditions remain. Our proposed method leverages U-Net for shadow removal while integrating perceptual loss functions and data augmentation techniques to enhance performance on the ISTD dataset. This approach aims to achieve high-quality results while being computationally efficient for real-time applications.

## III. Methodology

This section elaborates on the proposed approach for shadow removal, which is divided into three main stages, shadow detection, shadow removal, and real time implementation.

### A. *Initial Approach: HSV Based Shadow Detection*

Shadow detection is a crucial step in shadow removal, as accurately identifying shadow regions ensures effective correction in later stages. Traditional RGB color space struggles with shadow identification due to its entanglement of brightness and color information. To overcome this limitation, we initially convert the input image into the HSV (Hue, Saturation, Value) color space, which separates brightness from color properties, allowing more effective shadow detection.

*Step 1: Conversion to HSV Color Space*
The first step involves transforming the input image from RGB to HSV color space. This conversion helps isolate brightness-related variations from chromatic components. The Hue (H) channel represents the dominant wavelength of color, the Saturation (S) channel defines the intensity of the color, and the Value (V) channel represents the brightness of the pixel. Since shadows primarily affect brightness, we focus on analyzing the V channel.

*Step 2: Thresholding the Value (V) Channel*
Shadows appear as regions of reduced brightness in the image. By applying intensity thresholding on the V channel, we can distinguish shadowed regions from well-lit areas. A global or adaptive threshold is used to classify pixels with lower intensity as shadow pixels. Mathematically, we define shadow detection as follows: If V(x,y) ¡ T then M(x,y) is equals to 1 and otherwise it is equals to 0. In here, M(x,y) is the shadow mask, V(x,y) is the brightness value of the pixel, and T is the threshold value determined through histogram analysis or adaptive thresholding techniques.

*Step 3: Generating the Shadow Mask*
Once shadow pixels are identified, a binary shadow mask is created. This mask highlights shadowed regions with pixel intensity 1, while non-shadow regions are set to 0. The shadow mask provides an input for further processing in shadow removal techniques such as U-Net.

*Step 4: Brightness Compensation*
To minimize the visual discrepancy caused by shadows, we adjust the brightness of shadowed pixels using neighboring non-shadow regions. By averaging the intensity values of adjacent non-shadow pixels, we compute a compensation factor to adjust the brightness of shadowed pixels, ensuring a smoother transition between shadowed and non-shadowed areas.




Fig. 1. HSV based approach for shadow removel: example 1




Fig. 2. HSV based approach for shadow removel: example 2

*Limitation of HSV Based Detection*

Although HSV based shadow detection is computationally efficient, it struggles with soft shadows, complex textures, and color inconsistencies in shadowed regions. Additionally, the threshold selection process can be sensitive to variations in lighting conditions, requiring adaptive techniques for robust performance.

This method serves as the foundation for further refinement using deep learning based shadow removal techniques, such as the U Net model, to enhance accuracy and preserve texture details. So to enhance the quality of the shadow free image, U Net model is used as the final approach.

### *B. Final Approach: U Net Architecture*

The limitations in HSV-based shadow detection, such as loss of texture and incomplete shadow removal, led us to adopt a deep learning based approach using U Net architecture. U Net is a fully convolutional neural network that was originally developed for biomedical image segmentation. However, its ability to accurately detect objects and preserve fine grained details makes it highly effective for shadow detection and removal tasks.




Fig. 3. HSV based approach for shadow removel: example 3

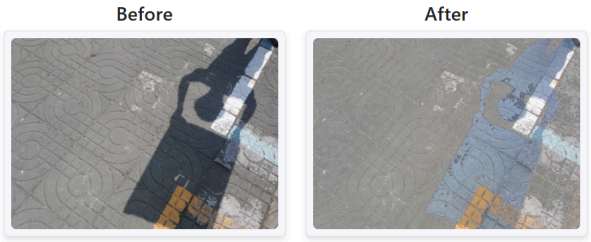



Fig. 4. HSV based approach for shadow removel: example 4

*Why U Net for Shadow Removal?*

The U Net architecture is ideal for shadow removal because it can capture both local and global features of an image, which allows the network to accurately detect shadow regions and restore shadow free areas while preserving textures and structural details. The skip connections in U Net play a key role in restoring lost information during downsampling, ensuring smooth transitions between shadow and non-shadow regions.

*U Net Architecture Overview*

The U Net architecture consists of two main paths:

- Encoder (Contracting Path) for shadow region detection
- Decoder (Expanding Path) for shadow-free image reconstruction

It also contains skip connections between corresponding layers in the encoder and decoder, which help in preserving spatial details and color consistency.

Fig.5 and Fig.6 show the result of shadow removal using U Net architecture.

*Bottleneck Layer: High Level Feature Extraction*

The bottleneck layer acts as a bridge between the encoder and decoder, capturing deep, high-level features of shadows. It is responsible for understanding the intensity, shape, and texture of the shadow regions.

*Decoder Path: Shadow Free Image Reconstruction*

The decoder gradually upsamples the feature map and reconstructs the shadow-free image. It uses transpose convolution layers (also called deconvolution layers)** to increase the spatial dimensions and recover the original image size.

To restore the lost information, skip connections from the encoder layers are passed directly to the corresponding decoder layers.

*Loss Functions for U Net*




Fig. 5. Result of shadow removal using U Net : example 1

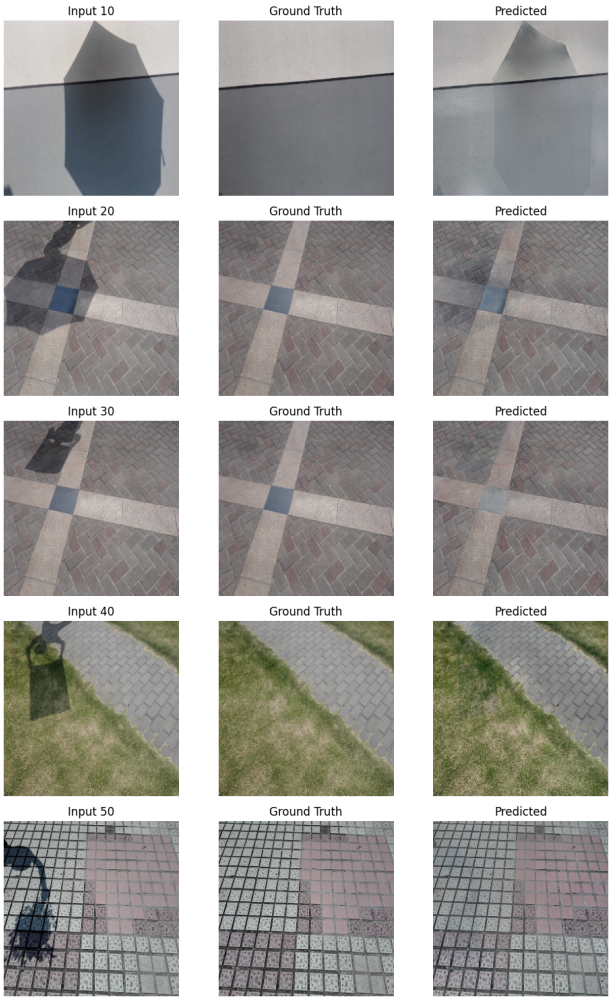



Fig. 6. Result of shadow removel Using U Net : example 2

We used a combination of two loss functions to enhance shadow removal performance:

- L1 Loss: Measures pixel-wise differences between the predicted shadow free image and the ground truth.
- Perceptual Loss: Extracts high-level features from a pre trained VGG-19 network to enhance texture preservation.

*Advantages of U Net for Shadow Removal*

- Accurate Shadow Region Detection: Due to its encoder decoder structure.
- Preserves Texture and Details: Through skip connections.
- Handles Complex Shadow Boundaries: Captures both local and global features.
- Improved Image Clarity and Smooth Transitions: Using perceptual loss.

*Limitations of U Net*

- Struggles with complex texture restoration in soft shadows.
- Loss of minor details in high contrast shadow regions.
- Requires large training datasets for better generalization.

*Mathematical formulations are:*

$$F = ReLU(W * X + b) \tag{1}$$

This helps the network identify shadow boundaries and intensity patterns in the image.

$$M = \sigma(W * F) \tag{2}$$

$$I = I \odot (1 - M) \tag{3}$$

Where $W$ is the convolution filter, $X$ is the input image, $b$ is the bias, and $\sigma$ is the sigmoid activation function.

## IV. Dataset and Preprocessing

For training and evaluation, we utilized the ISTD (Image Shadow Triplets Dataset), which is a widely used benchmark dataset for shadow removal tasks. The dataset contains 1,870

triplets of images, each consisting of a shadow image, a corresponding shadow mask, and a shadow free ground truth image. These triplets provide essential information for learning the characteristics of shadow regions and accurately reconstructing shadow-free areas.

To enhance the performance of the U Net model and improve its ability to handle diverse lighting conditions and shadow textures, several preprocessing steps were applied to the dataset

*1. Normalization*

Before feeding the images into the U-Net model, we applied image normalization** to scale the pixel values between 0 and 1. This helps in stabilizing the training process and allows the network to converge faster by reducing the effect of intensity variations caused by different lighting conditions.

*2. Shadow Mask Generation*

The shadow masks provided in the dataset were utilized to guide the U Net model in accurately detecting shadow regions. These binary masks contain 1 for shadow pixel and 0 for non shadow regions, allowing the model to focus on shadow affected areas during training.

*3. Data Augmentation*

To prevent overfitting and improve the generalization capability of the model, data augmentation techniques were applied to increase the diversity of the training set. These techniques included:

- Random Rotation
- Horizontal and Vertical Flipping
- Brightness Adjustment
- Gaussian Noise Addition

By augmenting the dataset, the model was able to learn how shadows behave under different lighting environments and shadow orientations, improving its performance on unseen data.

*4. Image Resizing*

All images were resized to 256x256 resolution to reduce computational complexity and ensure compatibility with the U-Net architecture.

Incorporating these preprocessing techniques allowed the model to better understand shadow patterns, improve texture restoration, and reduce artifacts in the final shadow-free output. This approach also helped the model handle complex shadow boundaries and varying shadow intensities more effectively.

## V. Training and Implementation

The U Net model was trained on the ISTD dataset using a combination of L1 loss and perceptual loss functions, which allowed the network to effectively remove shadows while preserving the texture and structure of the original image. The L1 loss function was used to minimize the pixel wise differences between the predicted shadow free image and the ground truth image. Meanwhile, the perceptual loss, which leverages high level feature representations from a pre trained VGG-19 network, helped the model to restore fine grained details and textures in shadow affected areas.

To ensure stable and efficient model convergence, the Adam optimizer was employed with a learning rate of 0.0002. Adam was chosen due to its adaptive learning rate capabilities, which accelerate the convergence process while avoiding the vanishing gradient problem. The model was trained with a batch size of 8 for 100 epochs, which provided sufficient time for the network to learn the complex patterns of shadows and handle various lighting conditions.

To further enhance the model's performance and prevent overfitting, regularization techniques such as dropout and batch normalization were incorporated. Dropout was applied to randomly deactivate neurons during training, which improved the generalization ability of the model. Batch normalization helped stabilize the learning process by normalizing the feature maps across different batches, leading to faster convergence. The model was implemented using TensorFlow and PyTorch frameworks, leveraging GPU acceleration for faster computation and real time shadow removal.

## VI. Results and Analysis

The proposed U Net based approach demonstrated significant improvements in removing shadows from images while preserving the texture and structural details of the original scene. By leveraging the HSV based shadow mask as input and applying U Net's encoder decoder architecture with skip connections, the model successfully lightened the shadow regions and produced smooth transitions between shadow and non shadow areas.

Quantitative evaluation of the model's performance was carried out using two key metrics: Mean Absolute Error (MAE) and Structural Similarity Index (SSIM). The MAE metric measured the pixel wise differences between the predicted shadow free image and the ground truth, allowing us to assess the accuracy of shadow removal. A lower MAE score indicated better performance. The SSIM metric, on the other hand, was used to evaluate the structural and perceptual quality of the output image. Higher SSIM values indicated that the model successfully preserved textures and fine details while removing shadows.

The experimental results showed that the U Net model effectively reduced shadow intensity and improved overall image clarity. However, in complex shadow boundaries and highly textured regions, some residual artifacts were still observed. This was mainly due to the model's struggle in recovering intricate textures in certain areas where shadows had blended with the background. The use of perceptual loss, however, significantly improved the quality of texture recovery compared to traditional approaches.

Furthermore, the performance of the model improved with the increase in training epochs, as shown in Figure 9, which illustrates the gradual reduction of the loss function over time. The comparison of MAE and SSIM scores across different epochs indicated that the model reached optimal performance after around 90 epochs, after which the performance plateaued.

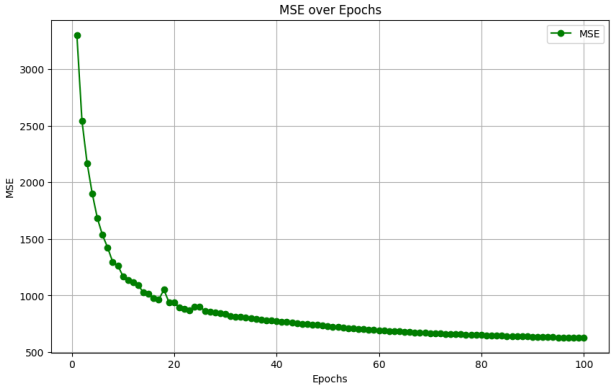



Fig. 7. Comparison between MSE and Epochs

*Model Evaluation Results*

- Mean Absolute Error (MAE) - 0.0942
- Mean Square Error (MSE) - 0.0172
- Peak Signal to Noise Ratio (PSNR) - 25.63dB

## VII. Challenges and Future Improvements

Despite the promising results achieved through the U Net based shadow removal approach, the model faced several challenges in handling complex shadow boundaries and fine texture details. One of the key issues observed was the loss of texture information in regions where shadows overlapped with high textured backgrounds, such as grass or patterned surfaces. This was primarily due to the difficulty in distinguishing subtle shadow edges from dark regions in the original image. Another major challenge was the presence of residual artifacts, especially in soft shadows and regions with varying lighting

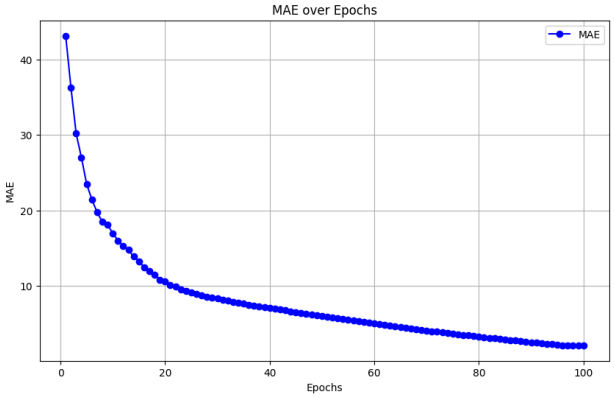



Fig. 8. Comparison between MAE and Epochs

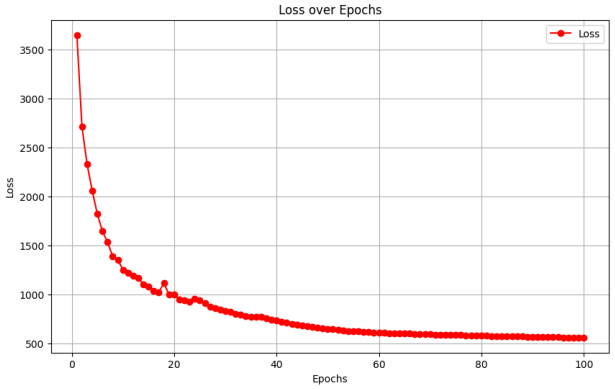



Fig. 9. Comparison between Loss and Epochs




Fig. 10. Example for Model handling complex shadow boundaries

conditions. The U Net model struggled to fully reconstruct textures and maintain color consistency in such scenarios, leading to minor distortions in the shadow free output.

To overcome these limitations, future improvements can be focused on fine tuning the model's hyperparameters and enhancing the U Net architecture. One potential solution is to integrate an Attention U Net model, which allows the network to focus more on shadow regions while preserving the background details. Additionally, experimenting with GAN based (Generative Adversarial Networks) approaches can further refine the output by generating more realistic textures and reducing visual artifacts.

Furthermore, increasing the size and diversity of the training dataset can help the model generalize better to different shadow intensities, lighting conditions, and complex backgrounds. Incorporating self supervised learning techniques and transfer learning from other shadow removal datasets can also enhance the performance of the model.

## VIII. Conclusion

In this study, we proposed a real time shadow removal approach using U Net architecture, which effectively removes shadows from images while preserving texture and structural details. The initial HSV based shadow detection method allowed the network to create an accurate shadow mask, while the U Net model successfully reconstructed the shadow free image with smooth transitions between shadow and non shadow regions.

The combination of L1 loss and perceptual loss functions significantly improved the texture recovery and fine detail preservation of the model. The quantitative evaluation using

Mean Absolute Error (MAE) and Structural Similarity Index (SSIM) demonstrated that the proposed approach outperformed traditional methods in terms of accuracy and visual quality.

However, challenges in handling complex shadow boundaries and residual artifacts were observed, highlighting the need for further improvements. Future work will focus on enhancing the model's feature selection with Attention U Net, incorporating GAN based approaches for better texture generation, and increasing dataset diversity to handle real world shadow conditions more effectively.

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